7. Development of the town

The trends of the future development of the town or city should be predicted and properly adjusted while planning the water supply scheme.

4.1.3 Water Demands

The first step in the design of a water supply project is the determination of water demand. This includes the quantity of water that will be required for various purposes with the provision for the estimated requirements of future. Next a reliable source of water must be located.

Following are the various types of demand,

1. Domestic demand
2. Industrial and commercial demand
3. Public demand
4. Fire demand
5. Loss and waste

1. Domestic demand

The domestic demand includes the water required in the houses for drinking, cooking, bathing, washing, sanitary purposes, private uses, gardening, etc. In Indian towns or cities, the domestic consumption of water under normal condition is 135 litres/day/capita as per IS: 1172-1993. This amount is about 50% of total consumption.

2. Industrial and Commercial Demand

This includes water required for hotels, factories, business centres, diary, sugar refineries, stores, offices, etc. The Consumption will vary greatly with the character of the city. This amounts to 20-25% of the total consumption.

3. Public Demand

This includes water required for flushing sewers, sprinkling streets, fountains, ornamental displays, swimming pools, public buildings, temples etc. This amounts to about 10% of the total consumption.

4. Fire Demand

It is the quantity of water required for fire fighting purposes. In India, the quantity of water required for fire prevention is taken as 1lit/capita/day. This amounts to 5 to 10% of total consumption.

5. Loss and waste

This includes the water loss in pipe line due to various reasons like defective pipe joints, cracks, thefts, faulty valves and fittings. This amounts to about 15% of total consumption.

4.1.4 Per Capita Demand

It is the annual average amount of daily water required by one person and includes the domestic use, Industrial and commercial use, public use, wastes, thefts etc. It may therefore be expressed as

Per capita demand = Q/(P*365) lits/day

Where Q = Total Quantity of water required by a town per year in litres

P = population

Factors affecting the per capita Demand

The following factors affect the per capita demand

1. Climatic conditions
2. Cost of water
3. Distribution pressure
4. Habits of people
5. Industries and commerce

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